

Activity 10.4: Verification Against Analytic Solutions

Geophysics — Numerical Heat Flow

Purpose

To demonstrate the importance of verifying numerical models against solutions with known behavior.

Concept

Analytic solutions provide benchmarks for testing numerical implementations.

Tasks

1. Identify an analytic solution discussed earlier in the course that can be reproduced numerically.
2. Describe what aspects of the solution should be compared (e.g., temperature profile shape, temporal evolution, heat flow).
3. Explain how grid spacing and time step affect numerical accuracy.
4. Discuss why a numerically stable solution is not necessarily accurate.

Reflection

Why is verification especially important when modeling complex geological systems?